

Scaling Spherical CNNs

Carlos Esteves¹, Jean-Jacques Slotine², Ameesh Makadia¹

¹ Google Research ² MIT



PROBLEM

Spherical CNNs compute convolutions in the **spectral domain** — far costlier than planar convolution — so applications stayed confined to small, low-resolution, low-capacity problems.

No spherical CNN existed at the scale of planar nets like VGG-19, blocking large scientific datasets.

MOTIVATION

- Molecular property prediction and weather forecasting are naturally **spherical and rotation-related** — a perfect fit for rotation-equivariant spherical CNNs.
- A molecule's properties are invariant to rotation and the atmosphere is a signal on a sphere — equivariance should be a decisive inductive bias.
- Yet the field was dominated by equivariant GNNs and transformers; prior spherical CNNs were stuck on tiny sets like QM7, 18× smaller than QM9.

To compete, models must scale to QM9 (134K molecules) and high-resolution ERA5 grids.

METHOD

- **Phase collapse nonlinearity:** the modulus $|x|$ collapses phase to restore rotation invariance without information loss.
- **Spectral batch norm, spectral pooling**, and an efficient residual block with a Fourier-space skip connection (Figure 2).
- **TPU-tuned JAX rewrite:** Fourier transforms run as dense matrix multiplies (DFT), not FFTs, across up to 32 TPUs.

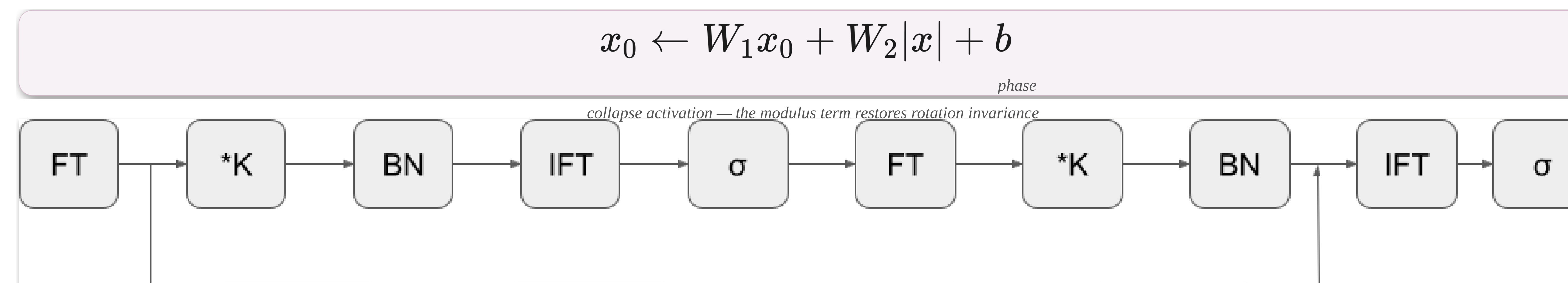
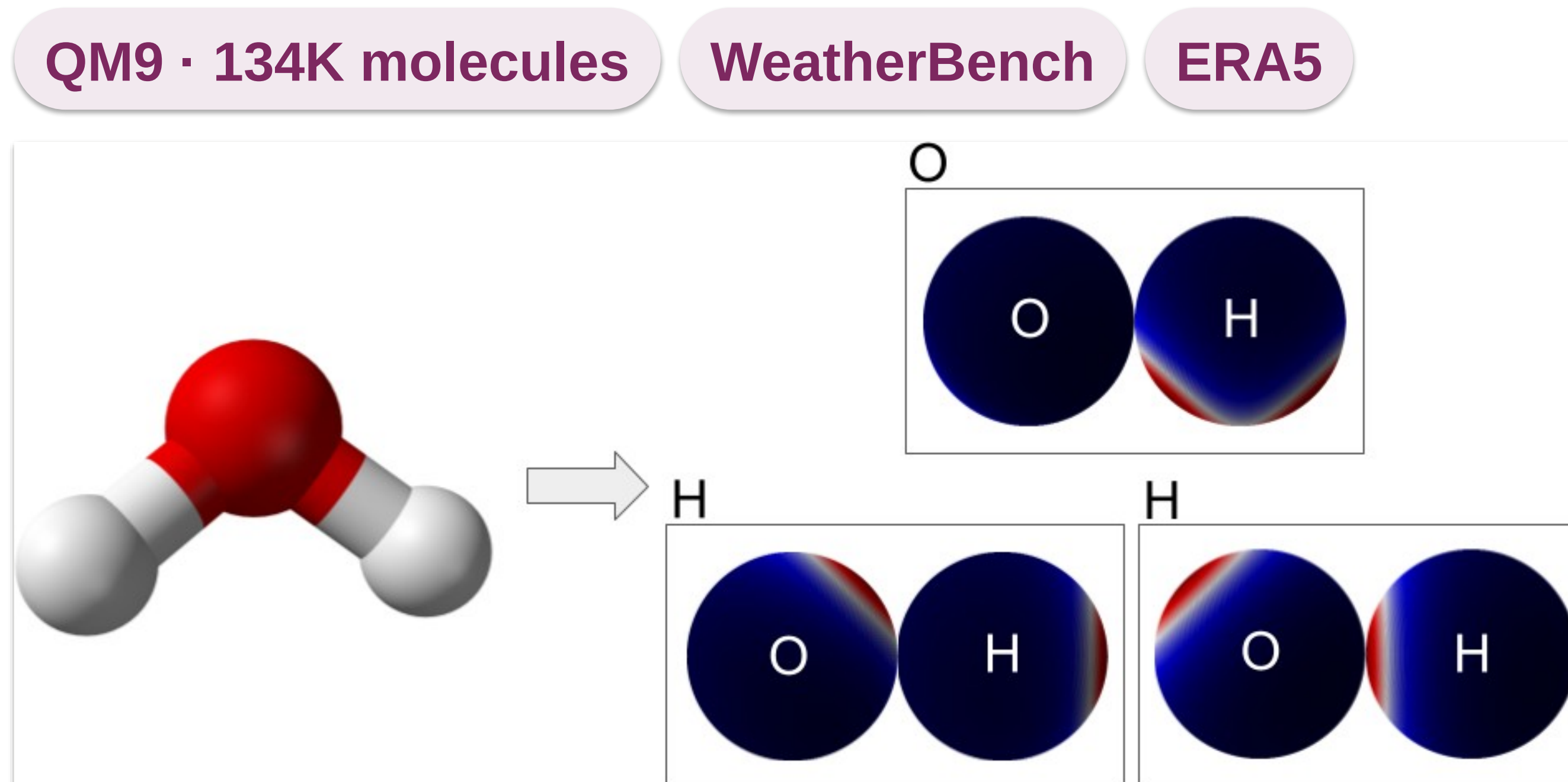


Figure 2. The efficient residual block: spin-weighted Fourier transforms (FT/IFT), filter multiply ($*K$), activation (σ), and spectral batch norm (BN). The skip connection happens in Fourier space.

DATASET / BENCHMARK



Molecules encoded as spherical functions per atom, from physically-based atom-pair interactions; QM9 regression + 3–28 day weather forecasting.

KEY RESULTS

QM9 (Split 1)	ϵ_{gap}	zpve	U_0	H
DimeNet++	32.6	1.21	6.32	6.53
TorchMD-Net	36.1	1.84	6.15	6.16
Ours	28.8	1.15	5.65	5.69

SOTA on 8/12 (Split 1) and 9/12 (Split 2) QM9 targets · lower is better (meV)

SO WHAT → Beats the previously dominant GNNs and transformers, and outperforms the WeatherBench baseline on all metrics in the 2-predictor setting.

ABLATION

COMPONENT	Δ STEPS/S	Δ RMSE
Phase collapse	−4.6%	−8.0%
Efficient residual	+19.3%	−2.4%

HEADLINE NUMBERS

10×

Larger spherical CNNs (ops & resolution)

8/12 QM9 SOTA targets (Split 1)

−8.0% RMSE from phase collapse

~100× faster on 32 TPUs

TAKEAWAY

Redesigned components plus a TPU-tuned JAX rewrite let spherical CNNs finally reach state-of-the-art molecular prediction.

The first demonstration that spherical CNNs are viable, competitive neural weather models.

SCAN TO READ

Paper & Code
arxiv.org/abs/2306.05420



Paper



Code